

Exploratory Data Analysis (EDA) Report

Cult Fit Booking Dataset

This report summarizes the structure, statistical characteristics, booking trends, high-demand activities, time patterns, and potential anomalies in the prepared Cult Fit dataset.

Dataset	Mentor_Cult_fit_Ritik.xlsx
Records	3,271
Columns	14
Key demand variable	Number_Booked
Key price variable	Price
Capacity variable	MaxBookees

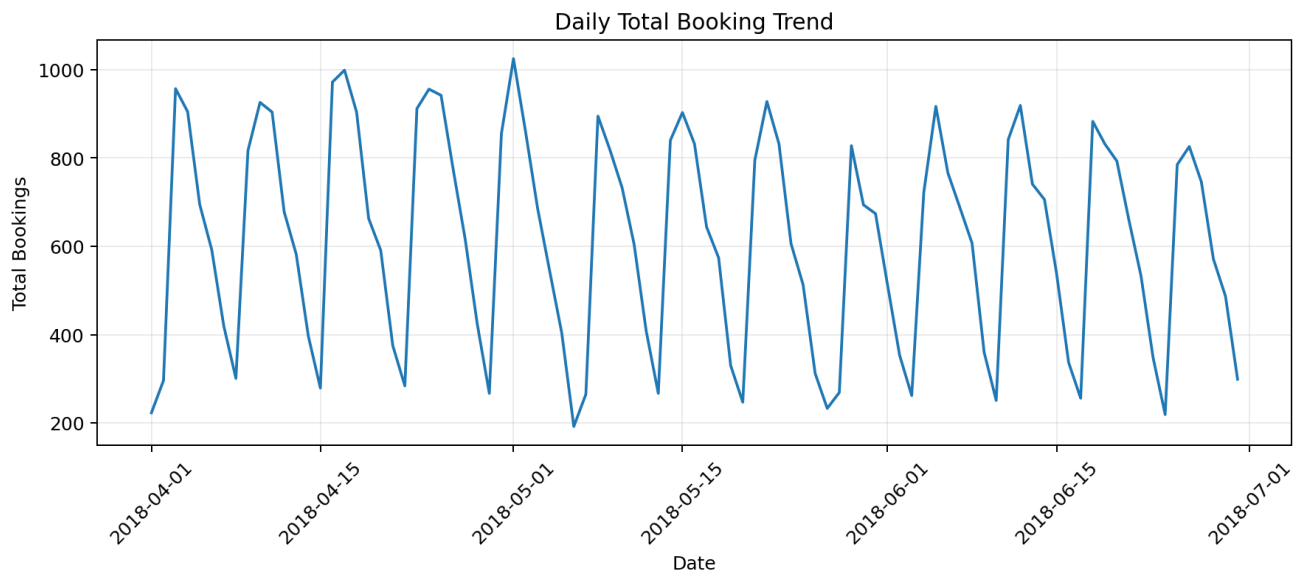
1. Data Overview

The prepared dataset contains **3,271 records** and **14 columns**. The analysis focuses on booking demand, price, capacity, class/activity, location/site, date, and class start time. The Excel index column was removed before analysis.

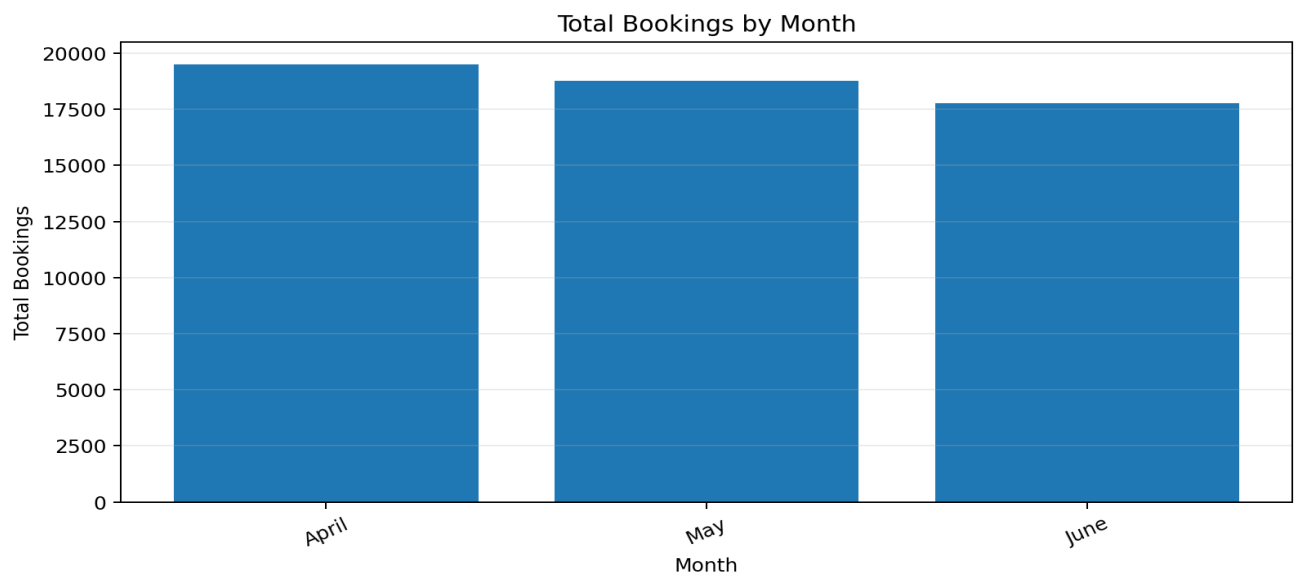
Variable	Mean	Median	Std. Dev.	Min	Max
MaxBookees	32.22	30.00	15.12	2.00	70.00
Number_Booked	17.13	16.00	9.52	1.00	64.00
Price	1852.71	1499.00	792.64	499.00	3999.00
Booking_Rate	57.23	55.56	31.91	1.67	516.67

2. Booking Trends and Time Patterns

Daily demand was created by aggregating Number_Booked by calendar date. This provides a clear view of changes in total booking activity over time.

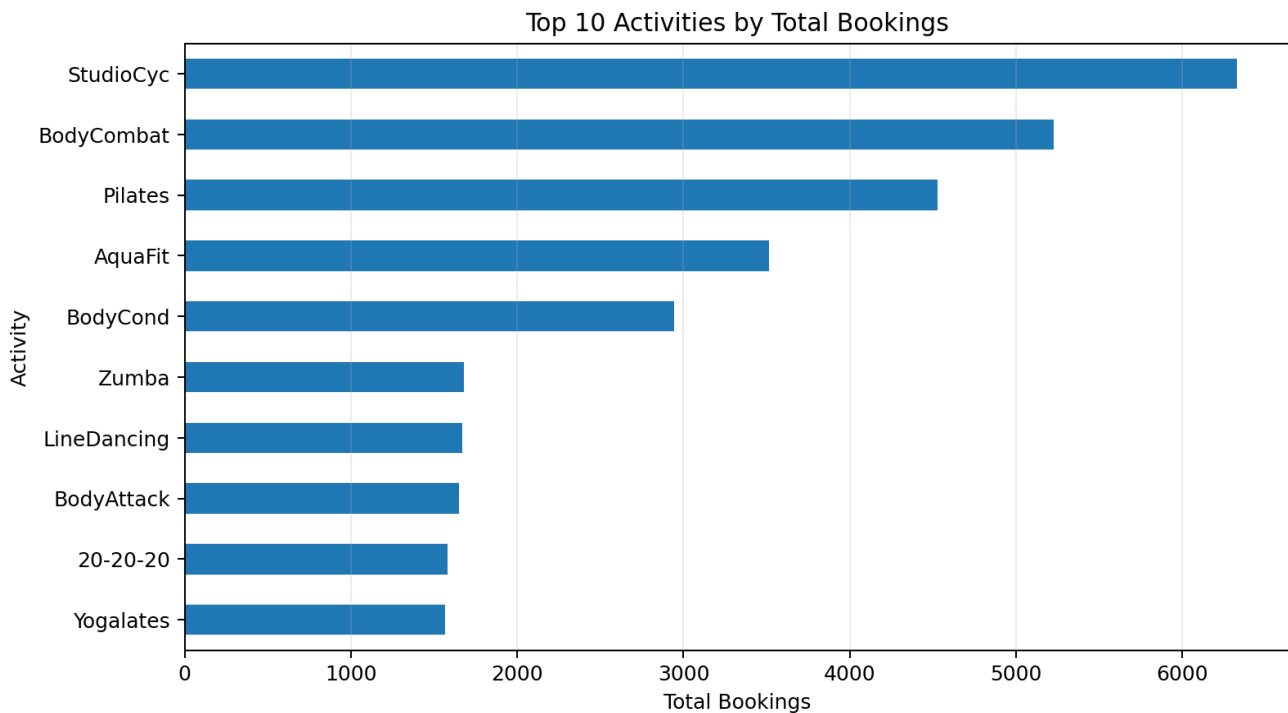


Monthly Demand



3. High-Demand Activities

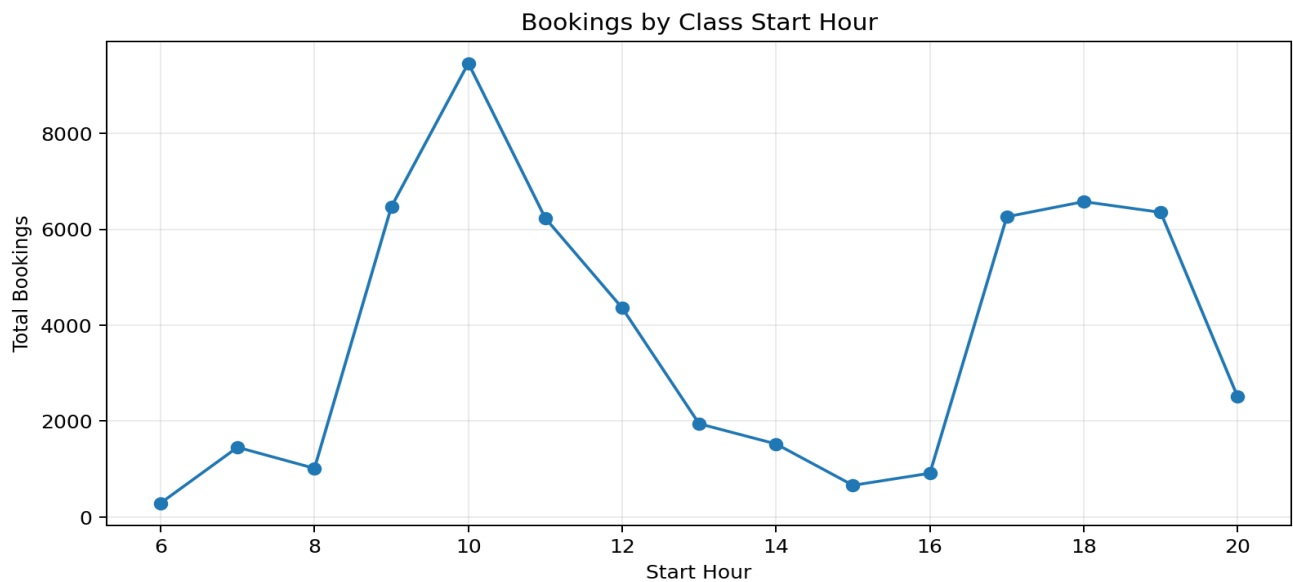
Activities were ranked by total Number_Booked across the dataset. This identifies classes that generated the greatest total booking volume.



Rank	Activity	Total Bookings
1	StudioCyc	6,330
2	BodyCombat	5,225
3	Pilates	4,527
4	AquaFit	3,514
5	BodyCond	2,942
6	Zumba	1,678
7	LineDancing	1,668
8	BodyAttack	1,648
9	20-20-20	1,582
10	Yogalates	1,568

4. Demand by Class Start Hour

Total bookings were grouped by class start hour to identify periods with higher booking activity.



The highest total booking volume occurs at start hour **10**, with approximately **9,466** bookings in the dataset.

5. Activity-Site Analysis

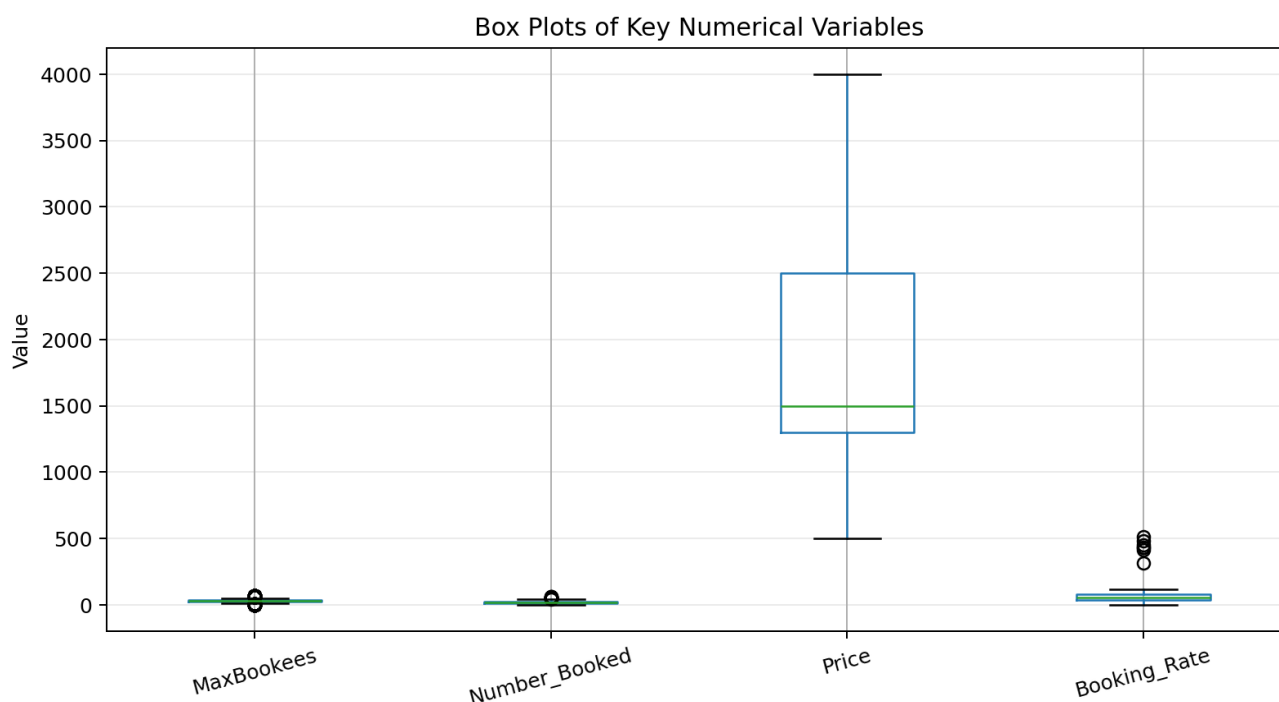
Activity Site	Total Bookings
BRP	14,068
HXP	12,391
SBP	12,297
NBL	10,793
TSC	6,470

Site-level differences indicate that location/activity-site should be considered when interpreting demand and when building predictive models.

6. Outlier and Anomaly Analysis

Potential numerical outliers were identified using the IQR rule: values below $Q1 - 1.5 \times IQR$ or above $Q3 + 1.5 \times IQR$. These observations should be investigated rather than automatically deleted.

Variable	IQR Outliers
MaxBookees	550
Number_Booked	17
Price	0
Booking_Rate	7



Capacity anomaly: 18 records have Number_Booked greater than MaxBookees. If MaxBookees represents the true booking capacity, these records may indicate data-entry, capacity-definition, or booking-system issues and should be reviewed.

7. Key Findings

1. The dataset contains 3,271 booking records across 14 columns.
2. Average Number_Booked is approximately 17.13 per record.
3. Average class capacity is approximately 32.22 where available.
4. Average listed price is approximately 1,852.71 where available.
5. April has the highest total booking volume among the months represented in the dataset.
6. StudioCyc has the highest total booking volume among individual activities.
7. Start hour 10 has the highest aggregate booking volume.
8. 18 records exceed stated capacity and warrant data-quality review.

8. EDA Implications for Further Modeling

1. Use date/time variables when building demand forecasting models.

2. Include activity/class and activity-site information because demand varies across these groups.
3. Investigate capacity anomalies before using booking utilization as a target or feature.
4. Consider weekly seasonality when forecasting daily demand.
5. For price elasticity modeling, distinguish price effects from differences between activity types and sites.
6. Retain genuine extreme observations where possible, but document any corrections or exclusions.

Prepared from the supplied Mentor_Cult_fit_Ritik.xlsx dataset. This report is an exploratory analysis and should be complemented by domain validation before business decisions are made.